

JACK’S RESEARCH LAB
Research Note 005

Did Remote Work Increase Productivity? Part II

Panel Evidence from U.S. Industries, 2015–2024

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Research Question: Once we compare industries to themselves over time, does rising remote work predict higher labor productivity?

Abstract

In Research Note 004, I found little evidence that industries with larger increases in remote work experienced faster productivity growth between 2019 and 2023. This follow-up note extends the analysis using a panel dataset constructed from approximately 29.4 million American Community Survey observations and Bureau of Labor Statistics industry productivity data.

The final dataset covers 49 matched three-digit NAICS industries from 2015 to 2024, excluding 2020 ACS data. I estimate pooled OLS models, industry fixed effects models, industry-and-year fixed effects models, and first-difference specifications.

The central result is clear: remote-heavy industries tend to look more productive in simple comparisons, but the relationship disappears once industry and year fixed effects are included. The evidence suggests that the apparent productivity advantage of remote work is mostly compositional. Remote work was concentrated in industries that were already different, rather than clearly causing measurable productivity gains within industries over time.

Study Design

Research Question	Did rising remote work increase productivity within industries?
Period	2015–2024, excluding 2020 ACS
Unit of Analysis	Industry-Year, three-digit NAICS
ACS Sample	29.4 million individual observations
Matched Industries	49
Final Panel	419 industry-year observations
Remote Work Measure	Share of workers reporting “worked at home”
Productivity Measure	BLS labor productivity index
Methods	OLS, fixed effects, first differences

Remote work is measured using the ACS transportation-to-work variable, where workers reporting “worked at home” are coded as remote workers. Education is measured as the share of employed workers with a bachelor’s degree or higher. Productivity is measured using the BLS labor productivity index for detailed industries.

The main panel model is:

$$\log(Productivity_{it}) = \alpha_i + \delta_t + \beta RemoteWork_{it} + \epsilon_{it}$$

where α_i represents industry fixed effects and δ_t represents year fixed effects. This specification asks whether increases in remote work within the same industry are associated with higher productivity, after controlling for economy-wide year shocks.

Key Finding

Key Finding: Remote work looks positively related to productivity in simple models, but the relationship disappears once industry and year fixed effects are included. The strongest specification estimates a remote-work coefficient near zero: -0.019 with a p-value of 0.963 .

This means the data provide little evidence that rising remote work increased productivity within industries over time. The positive relationship in simpler models appears to reflect differences between industries rather than productivity gains caused by remote work itself.

Main Results

The table below shows how the estimated remote-work coefficient changes across specifications.

Model	Remote Variable	Coefficient	p-value
Pooled OLS	WFH Share	0.418	0.006
Pooled OLS + College Share	WFH Share	0.700	<0.001
Industry Fixed Effects	WFH Share	0.614	0.011
Industry + Year Fixed Effects	WFH Share	-0.019	0.963
Industry + Year FE + College Share	WFH Share	0.374	0.237
First Differences	Change in WFH Share	0.018	0.919

The first three models suggest a positive relationship between remote work and productivity. But these specifications do not fully account for common year shocks and persistent industry differences. Once both industry and year fixed effects are included, the coefficient collapses toward zero and becomes statistically insignificant.

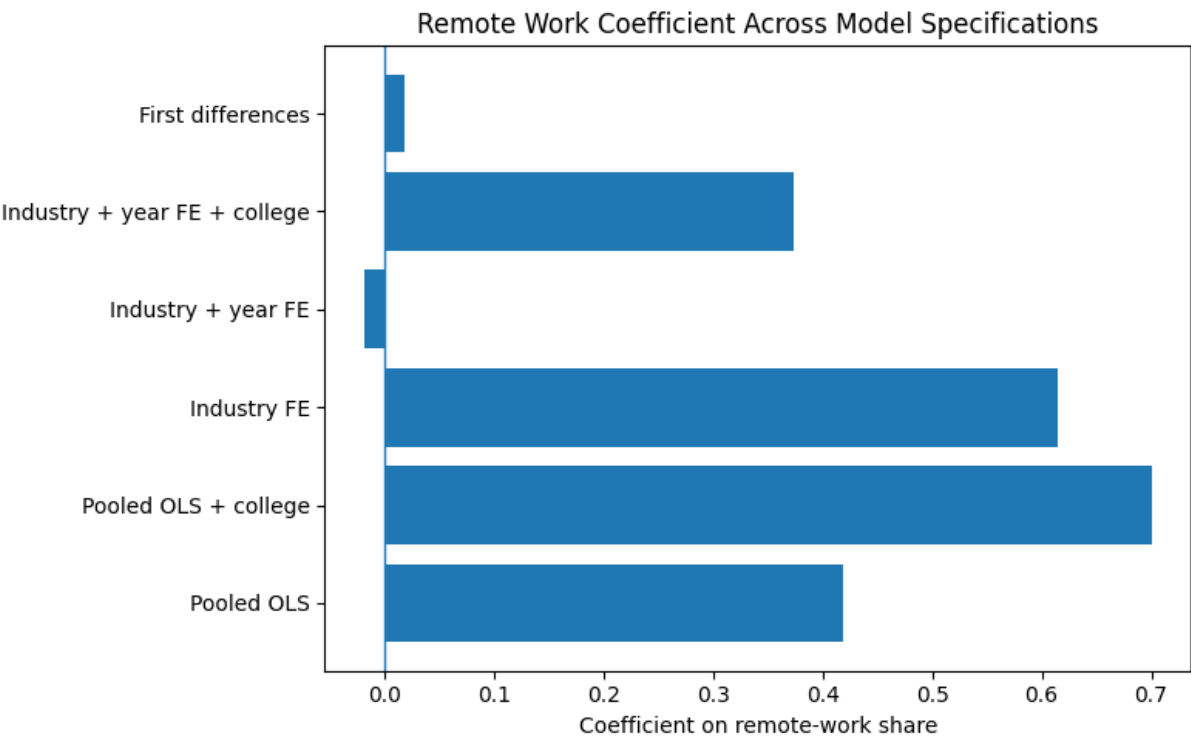


Figure 1: Remote Work Coefficient Across Model Specifications

This figure is the main result of the note. The positive relationship exists in simpler models, but it does not survive the stronger panel specifications.

Raw Relationship

The raw data show why the question is difficult. Some remote-heavy industries are highly productive, but the relationship is noisy and heavily shaped by industry composition.

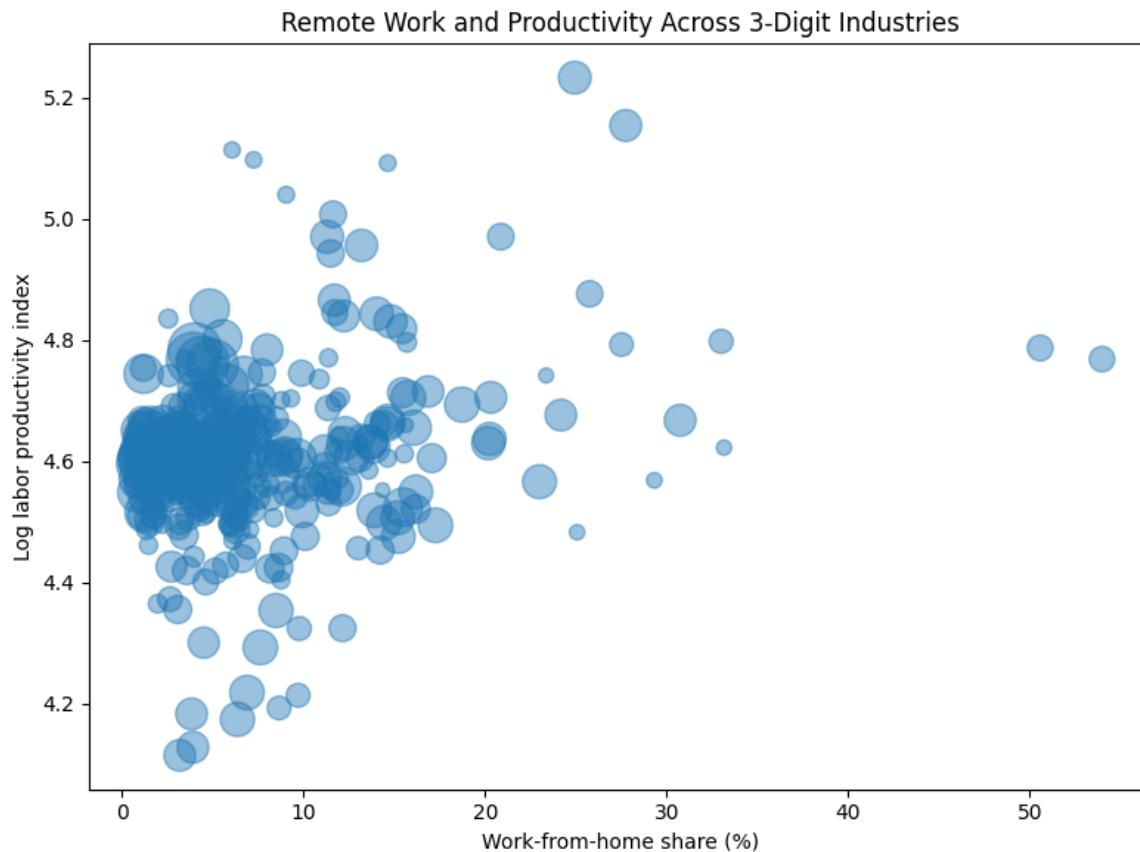


Figure 2: Remote Work and Productivity Across Three-Digit Industries

A simple scatterplot can make remote work look productive because remote-work industries are often knowledge-intensive, highly educated, and digitally suited. But that does not mean remote work itself caused the productivity difference.

Interpretation

The progression across models is more important than any single coefficient.

In pooled OLS, industries with higher remote-work shares also have higher measured productivity. This might initially look like evidence that remote work improved productivity. However, industries differ in many persistent ways. Software, finance, oil and gas, manufacturing, retail, and food service are structurally different industries.

The fixed effects models address this by comparing industries to themselves over time. In the industry-and-year fixed effects model, the question becomes:

[When an industry became more remote, did that same industry become more productive, net of year effects?]

The answer is no detectable relationship. The coefficient is approximately zero and the p-value is 0.963. The first-difference model reaches the same conclusion.

This does not prove that remote work has no effect for every worker, firm, or occupation. Instead, it suggests that at the industry level, increases in remote work were not clearly associated with measurable productivity gains.

What Changed from Research Note 004?

Research Note 004 compared productivity growth from 2019 to 2023 across industries. That was useful, but it was mostly cross-sectional. This note improves the analysis in three ways:

- It expands the ACS sample to 2015–2024.
- It uses 49 matched three-digit industries rather than a smaller cross-section.
- It estimates panel models with industry and year fixed effects.

The conclusion becomes stronger. The original note suggested that remote work was not clearly associated with faster productivity growth. This follow-up shows that the positive remote-work productivity relationship disappears under stronger panel controls.

Limitations

This analysis should still be interpreted as exploratory.

- The unit of analysis is the industry, not the firm or worker.
- ACS work-from-home status may not fully capture hybrid work.
- Productivity indexes reflect many factors beyond remote work, including capital investment, automation, demand shocks, supply chains, and energy prices.
- The panel is stronger than a simple cross-section, but it is not a randomized experiment.
- Remote work may have different effects across occupations, firms, and management systems that are hidden in industry-level averages.

A stronger causal study would use firm-level or worker-level data, better measures of hybrid work, and a source of quasi-random variation in remote-work adoption.

Data Sources

Remote-work and education measures were constructed from American Community Survey microdata accessed through IPUMS USA. The ACS sample covers 2015–2019 and 2021–2024. Remote work was identified using the transportation-to-work variable, where “worked at home” indicates working from home.

Labor productivity data were obtained from the Bureau of Labor Statistics Office of Productivity and Technology detailed industry productivity dataset. Productivity is measured using the labor productivity index for all workers at the three-digit NAICS industry level.

Conclusion

The headline result is simple: remote-heavy industries look more productive, but rising remote work does not clearly predict rising productivity within industries over time.

The positive relationship in simpler models appears to be driven by industry composition. Remote work expanded most in industries that were already more digital, more educated, and more suited to

knowledge work. Once those persistent industry differences and common year shocks are controlled for, the remote-work productivity relationship largely disappears.

The best interpretation is not that remote work obviously increased productivity, nor that it obviously reduced it. Instead, the industry-level evidence suggests a more cautious conclusion: remote work was concentrated in already-different industries, and its independent productivity effect is difficult to detect in aggregate productivity data.

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